

1. Webots

Installation

No	Source	location	desc	Actions
1	Webots	https://cyberbotics.com/	Download https://github.com/cyberbotics/webots/releases/download/R2025a/webots-R2025a.dmg	Dmg downloaded
2	mac	Local	Install	Installed dmg

Running

Applications/Webots.app/Contents/projects/robots/fujitsu/hoap2/worlds/hoap2_walk.wbt (hoap2) - Webots R2025a

Fujitsu HOAP-2 walk

0:00:10:250 - 1.01x

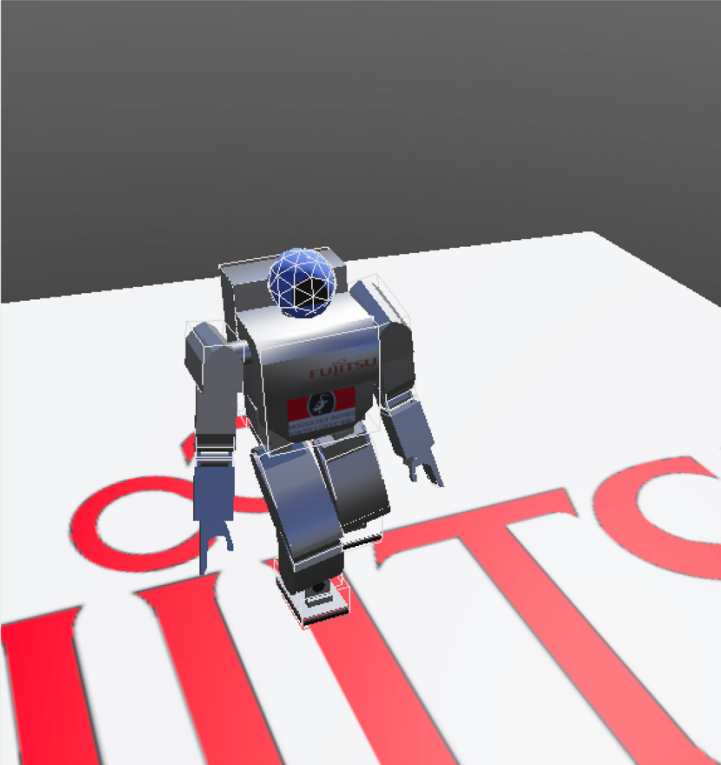
IMPORTABLE EXTERNPROTO

- WorldInfo
- Viewpoint
- TexturedBackground
- TexturedBackgroundLight
- DEF TATAMI Solid
- Hoap2 "HOAP-2"

Selection: Hoap2 (Robot)

Node	Position	Mass	Velocity
DEF:			

Print EXTERNPROTO



hoap2.c walk.csv

```
1 /*
2  * Copyright 1996-2024 Cyberbotics Ltd.
3  *
4  * Licensed under the Apache License, Version 2.0 (the "License";
5  * you may not use this file except in compliance with the License.
6  * You may obtain a copy of the License at
7  *
8  *     https://www.apache.org/licenses/LICENSE-2.0
9  *
10 * Unless required by applicable law or agreed to in writing, software
11 * distributed under the License is distributed on an "AS IS" basis,
12 * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either expressed or
13 * implied. See the License for the specific language governing permissions
14 * and limitations under the License.
15 */
16
17 /*
18  * Description: Reading a CSV file, getting motors positions,
19  *              CSV files are motion files used to control the robot.
20  */
21
22 #include <math.h>
23 #include <stdio.h>
24 #include <string.h>
25 #include <webots/emitter.h>
26 #include <webots/motor.h>
27 #include <webots/position_sensor.h>
28 #include <webots/robot.h>
29 #include <webots/touch_sensor.h>
30
31 typedef enum {
32     body_joint_1,
33     lleg_joint_1,
34     lleg_joint_3,
35     lleg_joint_2,
36     rleg_joint_1,
```

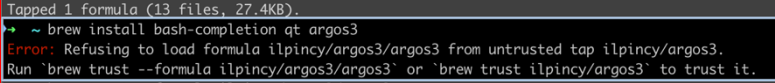
Console - All

```
Touch sensors: left force: 71.5 N right force: 0.0 N -> sum: 71.5 N
Touch sensors: left force: 67.3 N right force: 0.0 N -> sum: 67.3 N
Touch sensors: left force: 66.8 N right force: 0.0 N -> sum: 66.8 N
Touch sensors: left force: 66.0 N right force: 0.0 N -> sum: 66.0 N
Touch sensors: left force: 67.1 N right force: 0.0 N -> sum: 67.1 N
Touch sensors: left force: 67.2 N right force: 0.0 N -> sum: 67.2 N
Touch sensors: left force: 68.1 N right force: 0.0 N -> sum: 68.1 N
Touch sensors: left force: 0.0 N right force: 62.7 N -> sum: 62.7 N
Touch sensors: left force: 0.0 N right force: 68.0 N -> sum: 68.0 N
Touch sensors: left force: 0.0 N right force: 67.9 N -> sum: 67.9 N
Touch sensors: left force: 0.0 N right force: 68.4 N -> sum: 68.4 N
```

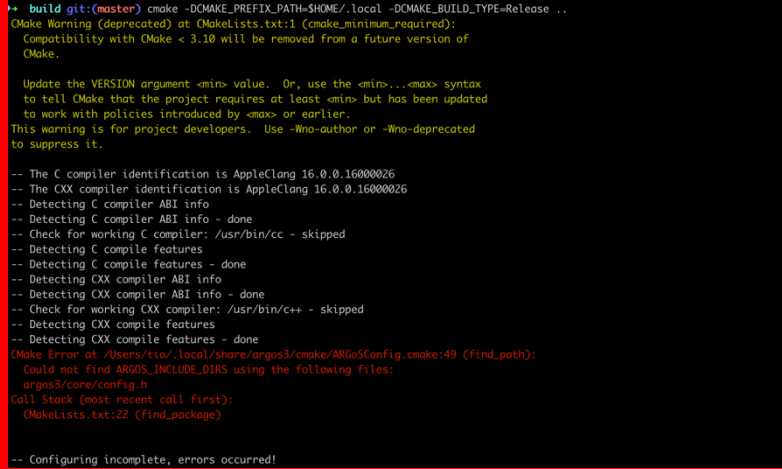
2. ARGoS 3

Installation

no	source	location	desc	actions	pass
1	ARGoS3	https://www.argos-sim.info/core.php	Find a guide	Homebrew Formula for MacOSX On MacOSX, ARGoS can be installed through Homebrew. Configuring Homebrew To get the Homebrew Formula working, you need to add the dedicated tap: <pre>\$ brew tap ilpincy/argos3</pre> This operation is necessary only once, to make the first installation of ARGoS possible. Installing ARGoS To install ARGoS for the first time, it is enough to write this command: <pre>\$ brew install bash-completion qt argos3</pre> This command installs ARGoS along with the recommended dependencies. Updating ARGoS To update an installed version of ARGoS to the newest version, first update the formulas: <pre>\$ brew update</pre> and then upgrade ARGoS: <pre>\$ brew upgrade argos3</pre>	1
2	Homebrew	mac	add ARGoS3 tap	<pre>brew update brew tap ilpincy/argos3</pre>	1
3	Homebrew	mac	add dependencies	<pre>brew install bash-completion qt</pre>	1

4	Homebrew	mac	Install ARGoS3	brew install argos3 	0
5	Homebrew	mac	Trust tap	brew trust ilpincy/argos3	1
6	Homebrew	mac	Install ARGoS3	brew install argos3 need < 3.5 but 4.4.3	0 cmake version
7	CMake	mac	cmake version	cmake --version: 4.4.3	1
8. brew failed so I gave up with Homebrew					
9	web	https://www.argos-sim.info/core.php	find a guide	ARGoS Source The ARGoS source is hosted on github: https://github.com/ilpincy/argos3 . To download it, open up a terminal and clone the repository with this command: <pre>\$ git clone https://github.com/ilpincy/argos3.git argos3</pre> To compile the source, follow these instructions .	1
10	git	https://github.com/ilpincy/argos3	check repo	<i>Mac OSX</i> On Mac, you can install all of the necessary requirements using HomeBrew . On the command line, type the following command: <pre>\$ brew install pkg-config cmake libpng freeimage lua qt \ docbook asciidoc graphviz doxygen</pre>	1
11	Homebrew	local	dependencies	brew install pkg-config cmake libpng freeimage lua qt docbook asciidoc graphviz doxygen	1
12	mac	local	folder	cd ~ mkdir temp	1

				cd temp	
13	git	local	git clone	git clone https://github.com/ilpincy/argos3.git argos3	1
14	Build	local	build directory	cd argos3 mkdir build_simulator && cd build_simulator	1
15	CMake	local	check cmake	cmake ../src - DCMAKE_POLICY_VERSION_MINIMUM=3.5	1
16	Make	local	make compile	make	1
17	CMake	local	local prefix configure try	cmake ../src - DCMAKE_POLICY_VERSION_MINIMUM=3.5 - DCMAKE_INSTALL_PREFIX=\$HOME/.local	1
18	Make	local	install	make install [100%] Built target prototype_radios_controller_autogen [100%] Built target prototype_radios_controller Install the project... -- Install configuration: "Release" -- Installing: /Users/tio/.local/bin/argos3 -- Installing: /Users/tio/.local/share/argos3/uninstall_argos3.sh -- Installing: /Users/tio/.local/include/argos3/core/config.h CMake Error at cmake_install.cmake:53 (file): file INSTALL cannot find "/Users/tio/Documents/temp/argos3/build_simulator/argos3.1.gz": No such file or directory. make: *** [install] Error 1	0 no argos3.1.gz
19	CMake	local	install with local path	cmake ../src \ -DCMAKE_POLICY_VERSION_MINIMUM=3.5 \ -DCMAKE_INSTALL_PREFIX=\$HOME/.local \ -DCMAKE_PREFIX_PATH=\$(brew --prefix qt)	1
20	Make	local	make install	make install	0 no doc build
21	Make	local	make doc install	make doc	1
22	Make	local	make install	make install	1
23	zsh	local	add path var	echo 'export PATH="\$HOME/.local/bin:\$PATH"' >> ~/.zshrc	1
24	zsh	local	apply	source ~/.zshrc	1
25	git	git clone https://github.com/ilpincy/argos3-examples	clone	cd ~ cd temp	1

				git clone https://github.com/ilpincy/argos3-examples.git	
26	CMake	local	build	<pre>cd ~/temp/argos3-examples mkdir build cd build cmake .. -DCMAKE_BUILD_TYPE=Release - DCMAKE_PREFIX_PATH=\$HOME/.</pre>  <pre>-- build git:(master) cmake -D CMAKE_PREFIX_PATH=\$HOME/.local -D CMAKE_BUILD_TYPE=Release .. OMake Warning (deprecated) at OMakeLists.txt:1 (cmake_minimum_required): Compatibility with OMake < 3.10 will be removed from a future version of OMake. Update the VERSION argument <min> value. Or, use the <min>...<max> syntax to tell OMake that the project requires at least <min> but has been updated to work with policies introduced by <max> or earlier. This warning is for project developers. Use -Wno-author or -Wno-deprecated to suppress it. -- The C compiler identification is AppleClang 16.0.0.16000026 -- The CXX compiler identification is AppleClang 16.0.0.16000026 -- Detecting C compiler ABI info -- Detecting C compiler ABI info - done -- Check for working C compiler: /usr/bin/cc - skipped -- Detecting C compile features -- Detecting C compile features - done -- Detecting CXX compiler ABI info -- Detecting CXX compiler ABI info - done -- Check for working CXX compiler: /usr/bin/c++ - skipped -- Detecting CXX compile features -- Detecting CXX compile features - done OMake Error at /Users/tio/.local/share/argos3/cmake/ARGoSConfig.cmake:49 (find_path): Could not find ARGOS_INCLUDE_DIRS using the following files: argos3/core/config.h Call Stack (most recent call first): OMakeLists.txt:22 (find_package) -- Configuring incomplete, errors occurred!</pre>	0 failed to find core library Could not find ARGOS_CMAKE_ DIR using the following files: FindARGoSQTOpenGL.cmake
27. I gave up using ~/.local path, but decided to use default path: /usr/local so I deleted all .local related data <pre>bash ~/.local/share/argos3/uninstall_argos3.sh</pre> <pre>rm -f ~/.local/bin/argos3 rm -rf ~/.local/include/argos3 rm -rf ~/.local/lib/argos3 rm -f ~/.local/lib/libargos3* rm -rf ~/.local/share/argos3</pre> <pre>source ~/.zshrc</pre> <pre>cd ~/temp/argos3</pre>					

```
rm -rf ./build-simulator
```

```
cd ~/temp/argos-examples
```

```
rm -rf ./build
```

```
cd ~
```

```
rm -rf
```

28	mac	local	create build folders	cd ~/temp/argos3 mkdir build-simulator cd ~/temp/argos3-examples mkdir build	1
29	CMake	local	default path build	cd ~/temp/argos3/ build-simulator cmake ../src \ -DCMAKE_POLICY_VERSION_MINIMUM=3.5 \ -DCMAKE_BUILD_TYPE=Release \ -DCMAKE_INSTALL_PREFIX=/usr/local \ -DCMAKE_PREFIX_PATH="\$(brew --prefix qt)"	1
30	Make	local	install doc	make doc	1
31	Make	local	install argos	sudo make -j"\$(sysctl -n hw.ncpu)"	1
32	mac	local	check	which argos3	1

check:

```
/usr/local/
```

```
|—— bin/
```

```
|  └── argos3
```

```
|—— include/
```

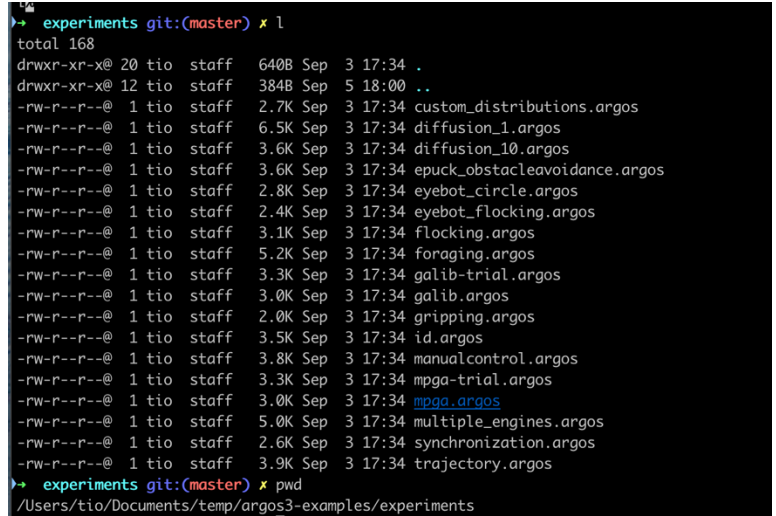
```
|  └── argos3/
```

```
|—— lib/
```

```
|  └── argos3/
```

```
└── share/
```

```
    └── argos3/
```

33	mac	local	argos3-exmaple	cd ~/temp/argos3-examples	1
34	CMake	local	build example	cmake -DCMAKE_BUILD_TYPE=Release ..	1
35	Make	local	install example	sudo make -j"\$(sysctl -n hw.ncpu)"	1
36	mac	local	check exmaple	cd ~/temp/argos3-examples cd experiments ls -al 	1
37	mac	local	run example	argos3 -c ~/temp/argos3-exmaples/experiments/diffusion_10.argos	1
38. Installation done and run an example successfully					

Running

